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SUPERSEDING
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MILITARY SPECIFICATION

ANTENNA AT-948/U

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers one type of UHF Antenna, the AT-948/U, a vertically polarized omnidirectional antenna having overhead coverage to fill in the cone of silence and operating over a frequency range of 400 through 550 megacycles for shipboard and shore station applications.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification, to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-15090	Enamel, Equipment, Light Gray (Formula No. 11)
MIL-E-16400	Electronic Equipment, Naval Ship and Shore: General Requirements for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-R-22732	Reliability Requirements for Shipboard and Ground Electronic Equipment
MIL-T-18303	Test Procedure; Preproduction and Inspection, (for Aircraft Electronic Equipment), Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment

Federal

TT-E-485	Enamel, Semi-Gloss, Rust Inhibiting

STANDARDS

Military

MIL-STD-242	Electronic Equipment Parts (Selected Standards)

FSC 5985

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2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts, Material and Processes - Material shall be in accordance with Specification MIL-E-16400, except as specified herein. In this regard, the selection of material shall be based on the fulfillment of the major design objectives as the prime consideration. Particular attention shall be paid to the acceptable and unacceptable material requirements of MIL-E-16400.

3.2 Design and Construction -

3.2.1 General - The equipment shall meet the requirements of Specification MIL-E-16400, except as specified herein. The design objectives of a high-strength-to-weight ratio, reliability, simplicity, ease of installation and maintenance consistent with the need of the Military service, shall be considered during all phases of equipment design and construction.

3.2.2 Service Conditions -

3.2.2.1 Temperature - The equipment shall meet the temperature requirements of MIL-E-16400 for Class 1 equipment.

3.2.2.2 Wind Load - The equipment shall be capable of withstanding a 120-knot wind.

3.2.2.3 Salt Atmosphere - The equipment shall be capable of operating without failure for a period normal to similar electronic equipment aboard ship, without being adversely affected by humid salt atmosphere.

3.2.2.4 Shock and Vibration - The complete antenna shall meet the applicable shock, vibration, and inclination requirements of Specification MIL-E-16400.

3.2.3 Construction - Equipment shall be constructed to conform to the service conditions as specified herein.

3.2.3.1 Mating Surfaces - Mating surfaces of dissimilar metals shall be protected to prevent corrosion according to the requirements of Specification MIL-E-16400.

3.2.3.2 Pressurization - The antenna shall be normally operated unpressurized. Design shall be such that when mounted vertically on a supporting structure, water or other liquids cannot collect to an extent that could ultimately result in failure or deterioration of performance of the antenna.

3.2.3.3 Maintenance and Repair - The antenna shall be so constructed that maintenance and repairs can be made with a minimum of disturbance to other components. Antennas shall be readily demountable and interchangeable to provide ease of servicing, maintenance and replacement.

3.2.3.4 Mounting Brackets - The antenna shall have, at its base, a mounting flange with bolt holes to anchor the antenna firmly in position when it is installed aboard ship.

3.2.3.5 Standardization - Every effort shall be made to standardize the units and parts entering into the construction, in conformance with MIL-STD-242, however, it is not intended to compromise the design or performance of the antenna produced as a result of this specification.

3.2.3.6 Tropicalization - Tropicalization will not be required.

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3.3 Performance Requirements -

3.3.1 Performance - The antenna shall be so manufactured that when mounted, in an unobstructed area, and sufficiently elevated above ground (or equivalent), the pattern of the antenna shall be such as to provide essential spatial coverage throughout a hemispherical volume.

3.3.2 Frequency - The antenna and feed system shall operate over the frequency range of 400 to 550 megacycles (mc).

3.3.3 Polarization - The antenna shall transmit r-f energy, vertically polarized in the horizontal field.

3.3.4 Gain - The gain of the antenna in the horizontal field shall be approximately five (5) decibels (db) above an isotropic (spherical radiation)source, or approximately three (3) db above a half-wave dipole source.

3.3.4.1 Vertical Gain - The gain of the antenna in the vertical (overhead) direction shall be approximately nine (9) db below an isotropic or approximately eleven (11) db below a half-wave dipole source. This is approximately fourteen (14) db below the maximum horizontal gain.

3.3.5 Power Rating (R-F) - The antenna shall be capable of handling 2000 watts of r-f power over the frequency range 400 to 550 megacycles (mc).

3.3.6 Standing Wave Ratio, Voltage - The impedance at the input to the antenna shall match the 50-ohm impedance of the input coaxial cable so that the VSWR does not exceed 1.8:1.0 over the entire 400 to 550 megacycle range.

3.3.7 Circularity - The antenna shall have a horizontal pattern that is omnidirectional to within ±1.0 db over the 400 to 550 megacycle frequency band.

3.3.8 Tilt - The antenna shall be designed to tilt the vertical lobe upwards to place the lower half-power point on or below the horizon.

3.4 Detail Requirements -

3.4.1 Functional Description - Antenna AT-948/U shall be a two element dipole collinear array housed within a polyester fiberglass vaportight cylindrical supporting structure. The dipole array shall produce omnidirectional vertically polarized energy and overhead coverage sufficient to fill in the cone of silence. The mounting flange shall be an integral part of the fiberglass watertight enclosure.

3.4.2 Overhead Coverage - To produce minimum interference with the rest of the pattern and optimum overhead coverage over the entire frequency region, the upper element of the dipole array shall be designed to provide orthogonal current distributions so as to fill in the cone of silence overhead.

3.4.3 Dimensions and Weight - The AT-948/U shall have a maximum height of 50 inches and a nominal diameter of 5-1/4 inches. The weight of the complete antenna shall not exceed 25 pounds.

3.4.4 Mounting Provisions - The mounting flange base of the AT-948/U Antenna shall be an integral part of the fiberglass weatherproof housing and shall be a flange 10 inches in diameter and approximately 1/2 inch thick with an 8-1/2 inch bolt circle on which there are eight (8) 7/8 inch diameter holes for mounting. The mounting flange shall mate with a standard class 125 type flange.

3.4.5 R-F Connectors - The r-f input to the antenna shall be by means of a connector capable of mating with connector UG-154A/U.

3.4.6 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.5 Workmanship - Workmanship shall be in accordance with Specification MIL-E-16400.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities of any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Equipments shall be subjected to the following tests to determine compliance with all the applicable requirements.

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include as a minimum the tests of 4.4.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the tests of 4.5 as a minimum.

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4.4 Preproduction Tests - The preproduction model antenna shall be subjected to the tests in the specification as follows, and conducted at a laboratory designated by the procuring activity.

4.4.1 Inclination - The antenna shall be subjected to inclination tests in accordance with MIL-E-16400.

4.4.2 Vibration - The antenna shall be subjected to vibration tests in accordance with MIL-E-16400.

4.4.3 Shock - The antenna shall be subjected to shock tests in accordance with MIL-E-16400.

4.5 Acceptance Tests - Each antenna produced shall be subjected to the acceptance tests detailed herein for acceptance before delivery. These tests are designed to insure that the complete antenna is in conformance with this specification.

4.5.1 Table of Tests - The tests to be performed shall be subdivided into two categories, (a) electrical, (b) mechanical.

4.5.1.1 Electrical Test No. 1 - Measurement of VSWR of complete antenna.

4.5.1.2 Mechanical Test No. 1 - Perform mechanical and visual inspection of complete antenna.

4.5.2 Test Procedure - Unless specified otherwise, all tests shall be performed under the following conditions:

Atmospheric Conditions - The tests shall be performed at prevailing factory or laboratory atmospheric conditions.

4.5.2.1 Electrical Tests - The antenna shall be tested in accordance with an approved test procedure.

4.6 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the procuring activity will furnish the procedures to be used for testing.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

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6. NOTES

6.1 Intended Use - The AT-948/U antenna is intended for use with drone command transmitting and receiving equipment in the frequency range from 400 to 550 megacycles (mc). The antenna is a general purpose antenna, however, and can be used for other applications within the limits of its frequency range, power, and gain characteristics. Primarily it is intended for installation in the upmost point aboard a ship.

6.2 This specification is not intended to be restricted with respect to the details of construction, except where such details are specified. Alternate forms of construction will be considered, provided that the contractor submits to the procuring activity a clear description for the points of difference of approval and provided that all performance requirements on this specification are met.

6.3 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over all referenced documents.

(2) This Specification - This specification shall have precedence over all applicable subsidiary documents. Deviations from this specification, or from subsidiary documents where applicable, may be made only upon written approval by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 9121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 49-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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